UNIVERSITÀ DI PARMA

SegnalER: progettazione e sviluppo di un motore di ricerca semantico per segnalazioni sanitarie

SegnalER: design and development of a semantic search engine for incident reports in health care

Alessandro Muzzi

SegnalER

Servizio regionale per l’incident reporting
in ambito sanitario

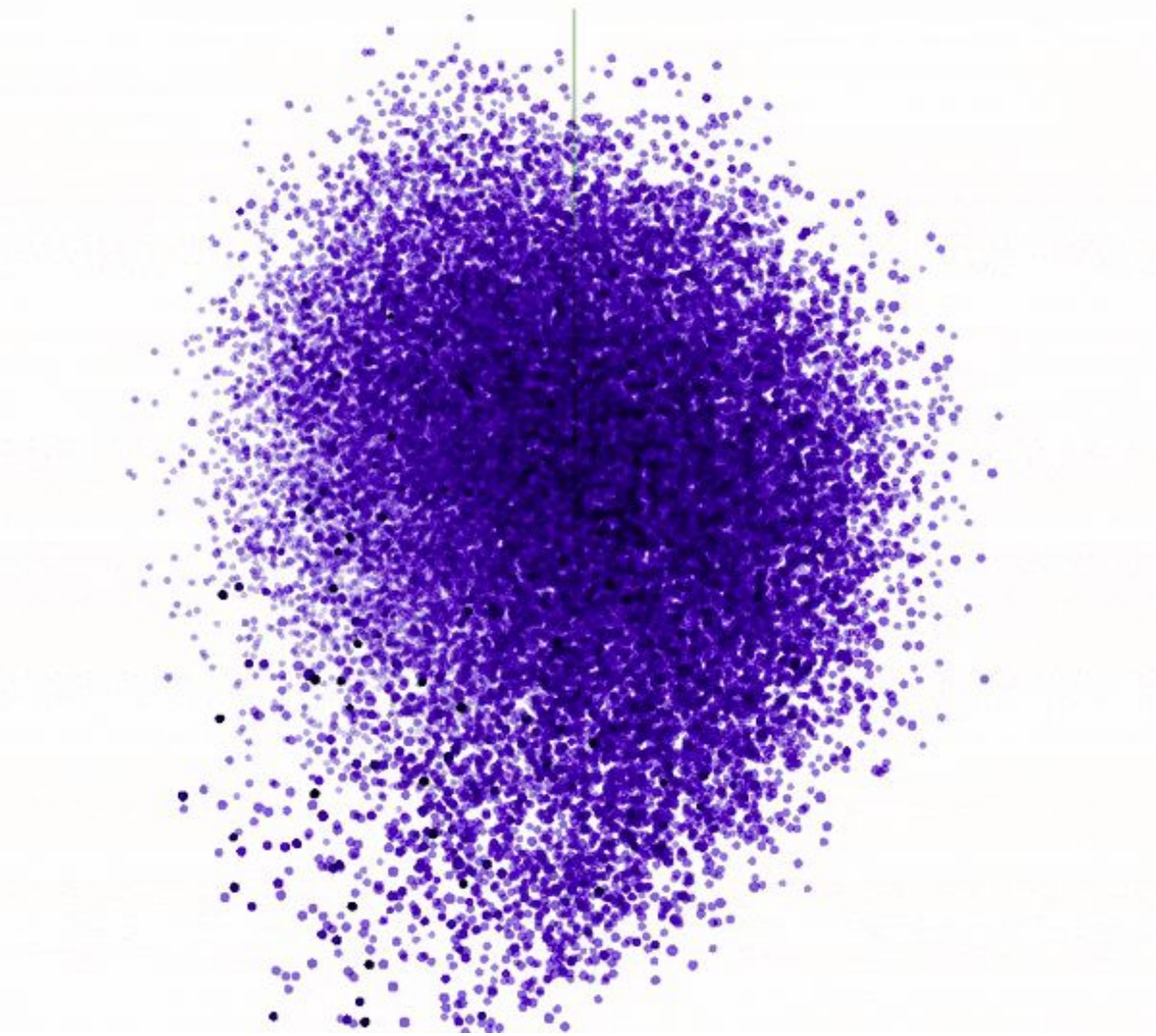
SegnalER

Semantic Search Engine

Text Processing Service

Correlation Engine

Dataset



IRDB

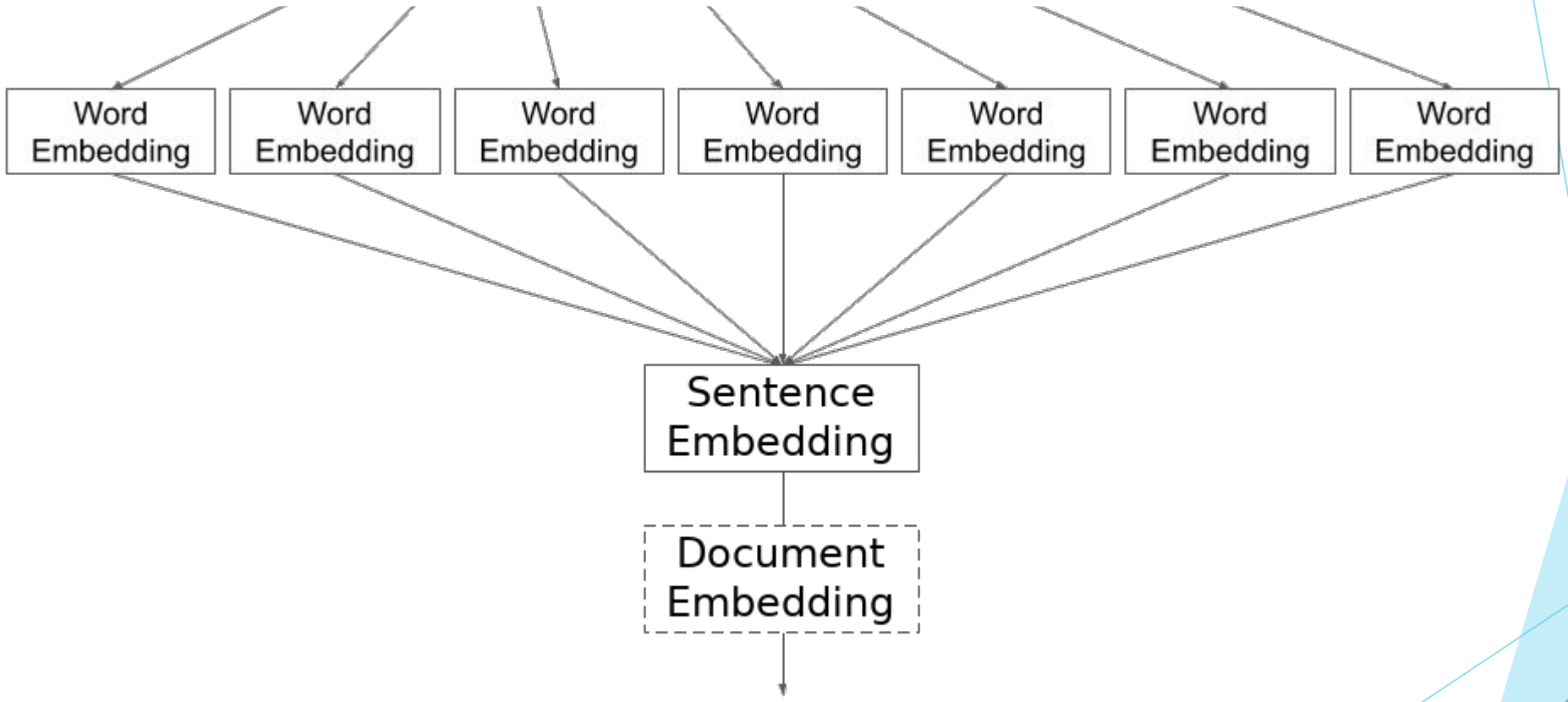
46.818 segnalazioni

3 problemi principali:

- dati sporchi
- scarsità di dati
- difficoltà ad interpretare gli embeddings

Embeddings

Scambio di siringa Somministrata efedrina anziché atropina



[0.23, 0.56, 0.67, 0.97, 0.05, 0.98, …, 0.13]

Embeddings

	Rappresentazione appresa	Embedding contestuali	OOV
Word2Vec	*Word*	✗	✗
GloVe	*Word*	✗	✗
FastText	*Subword*	✗	✓
BERT	*Subword*	✓	✓

Embeddings

Modelli Word Embedding

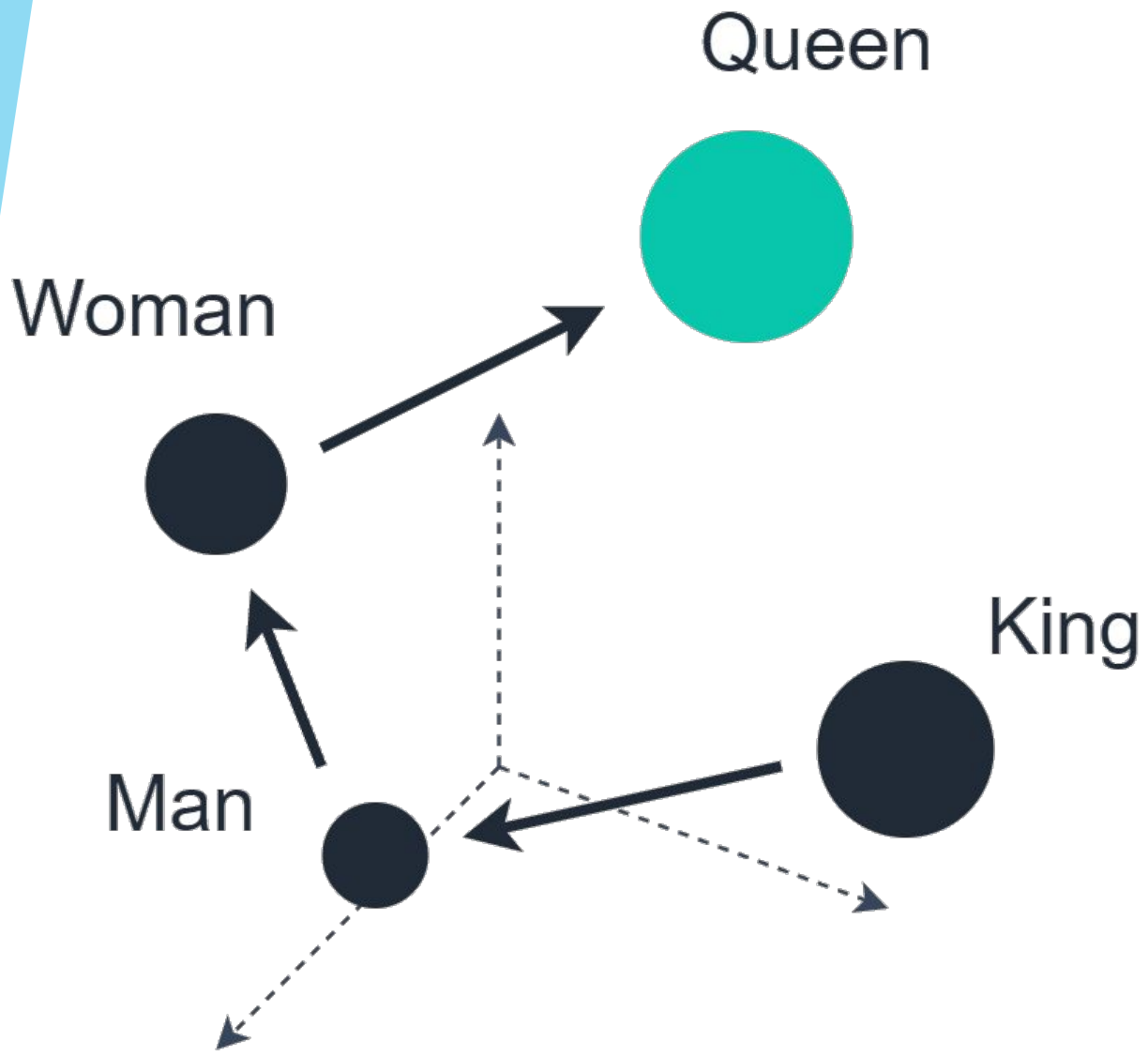
- Custom
- Wiki
- Wiki fine tuned custom

Tecniche Sentence Embedding

- Avg
- IDF
- SIF

9 modelli di Sentence Embedding

Metriche di confronto



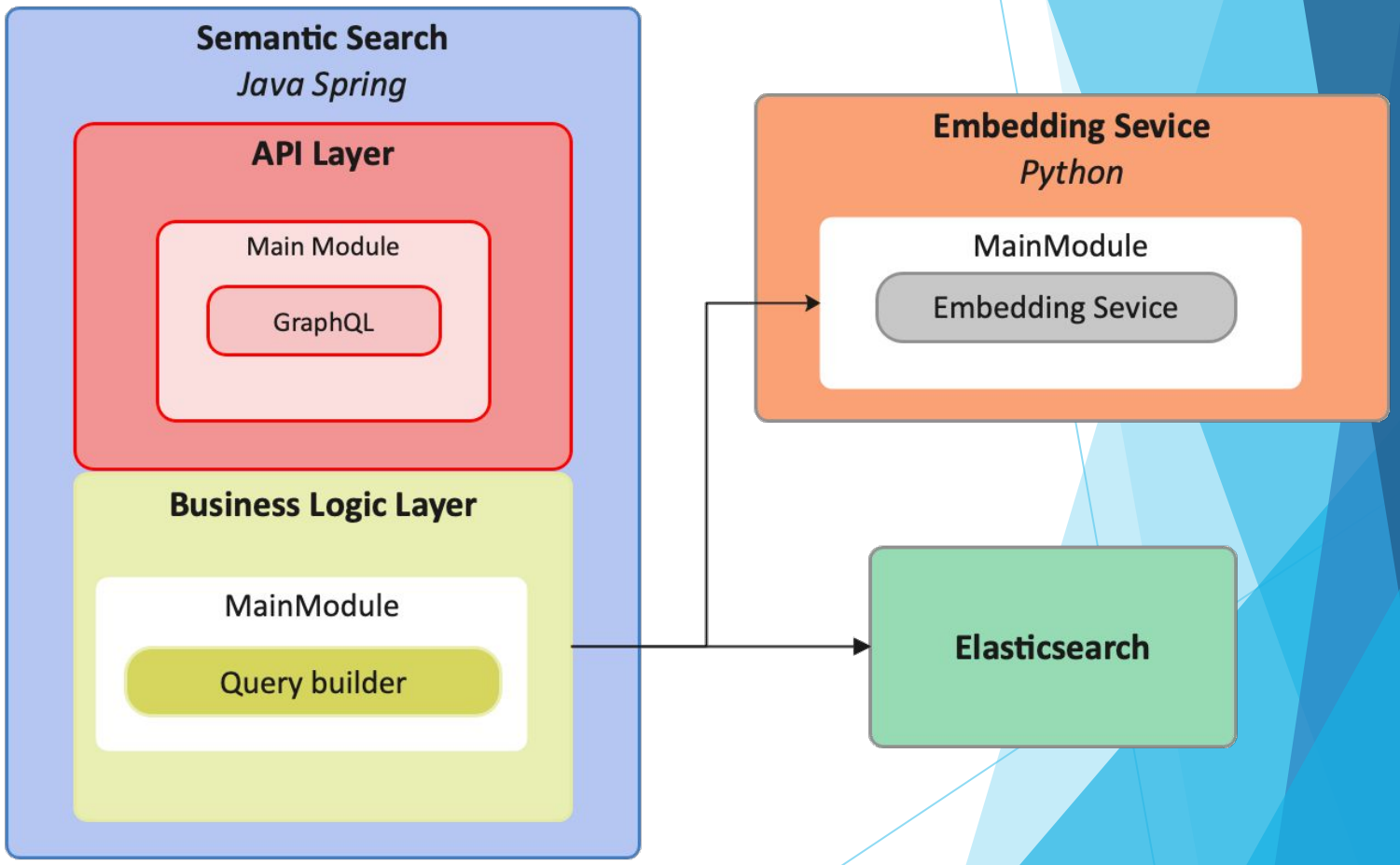
Euclidean distance

Cosine similarity

Ranking

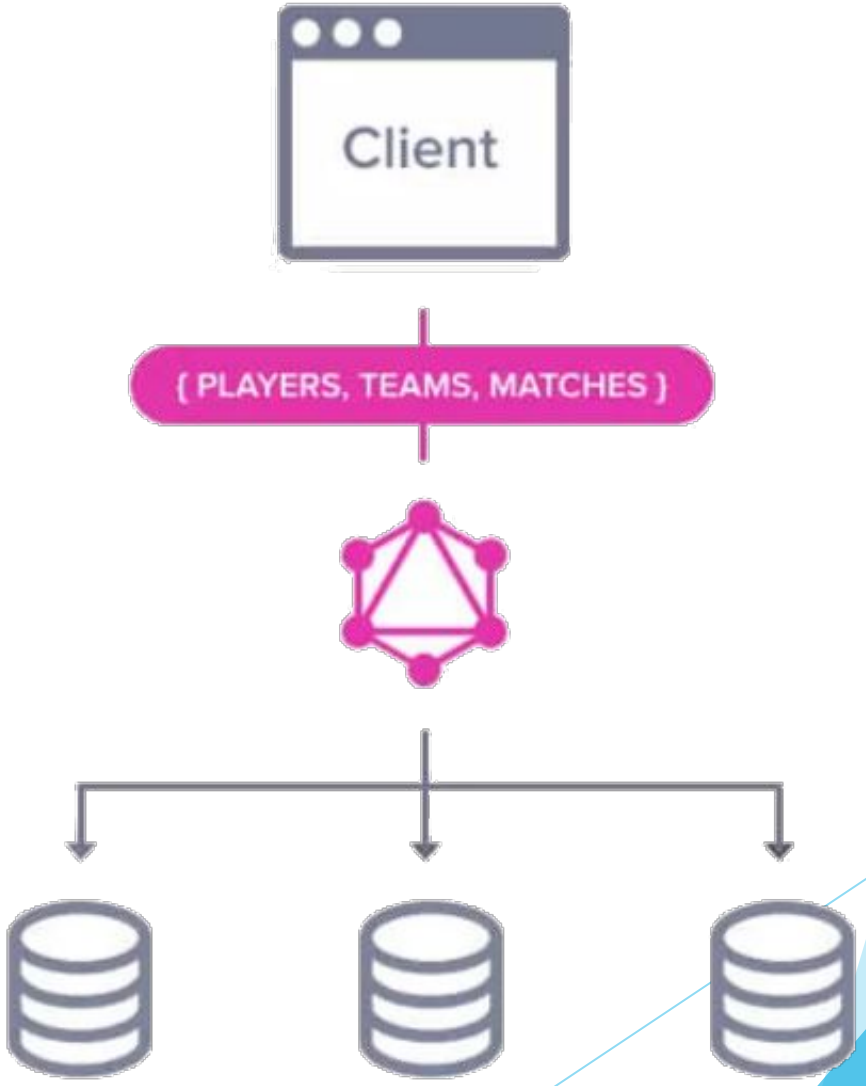
Architettura

- **Architettura modulare**
- GraphQL
- Elasticsearch



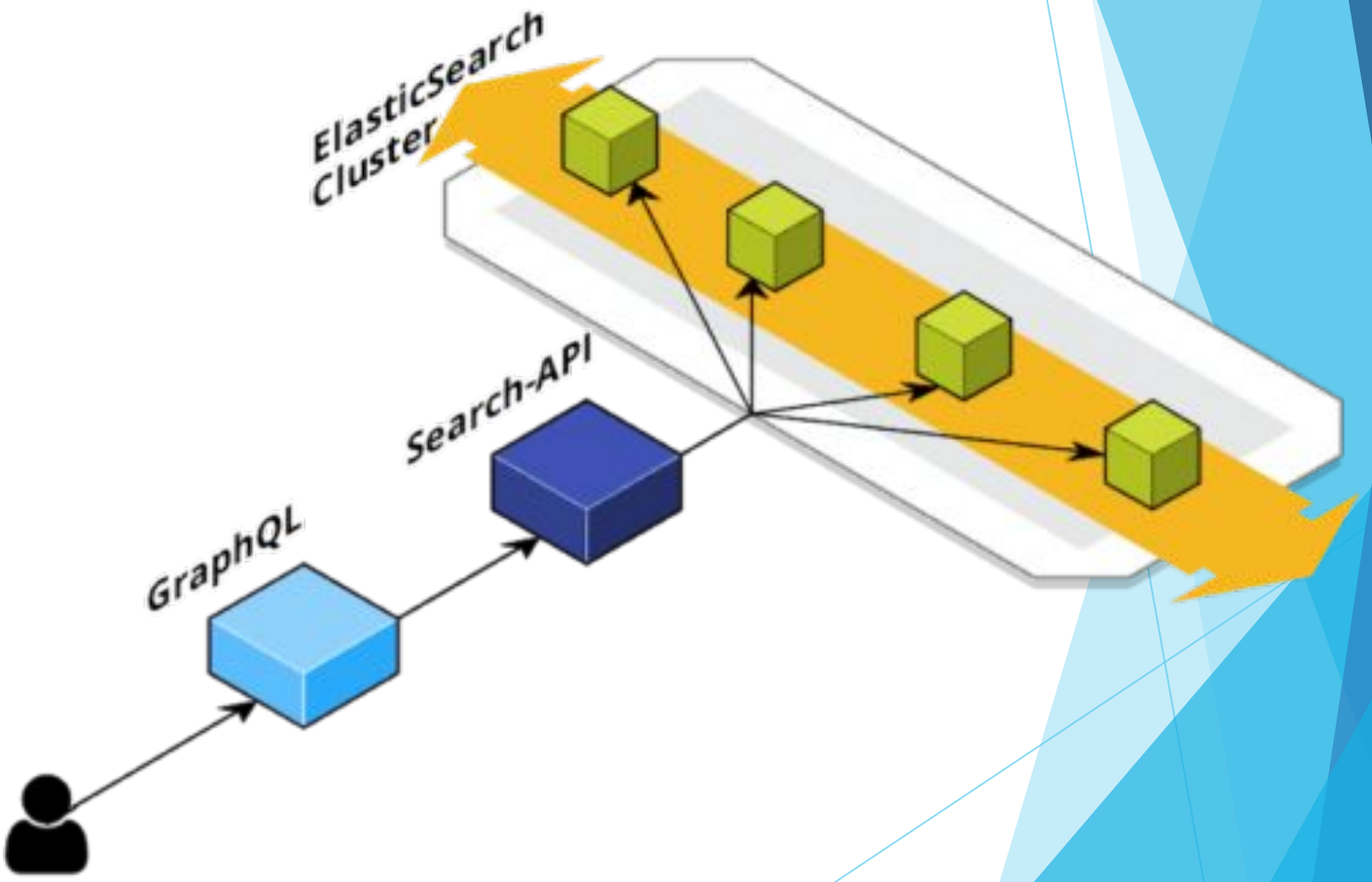
Architettura

- Architettura modulare
- **GraphQL**
- Elasticsearch

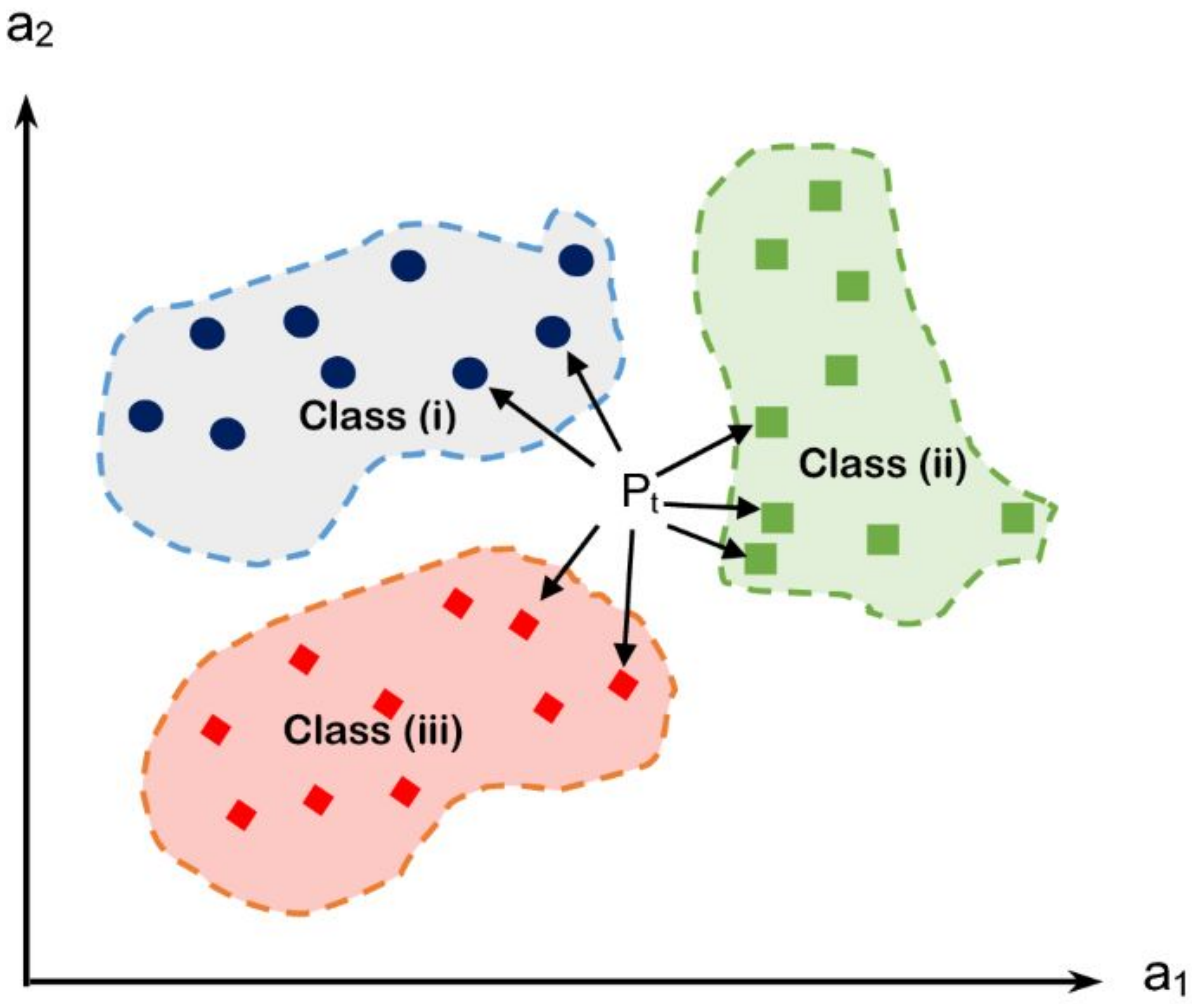


Architettura

- Architettura modulare
- GraphQL
- **Elasticsearch**

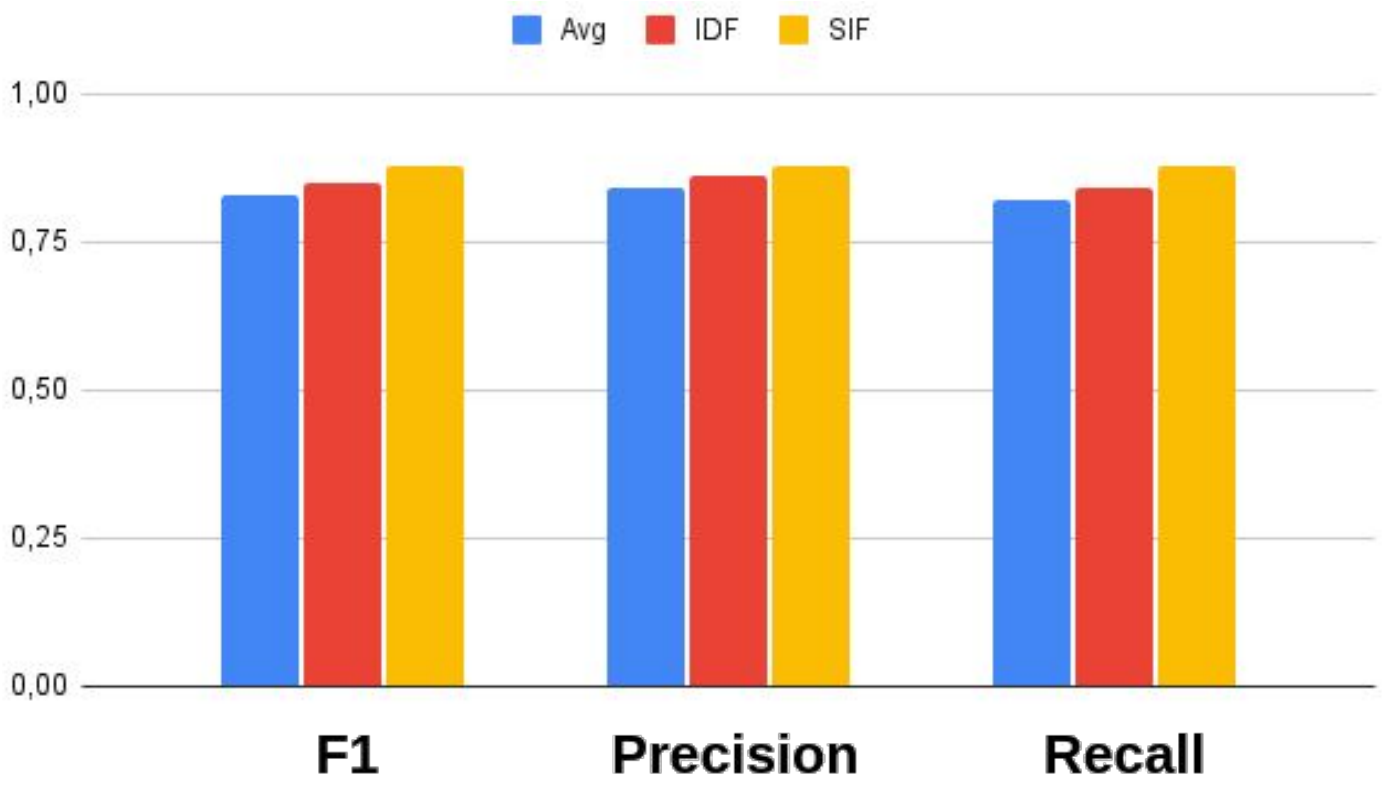


Valutazione



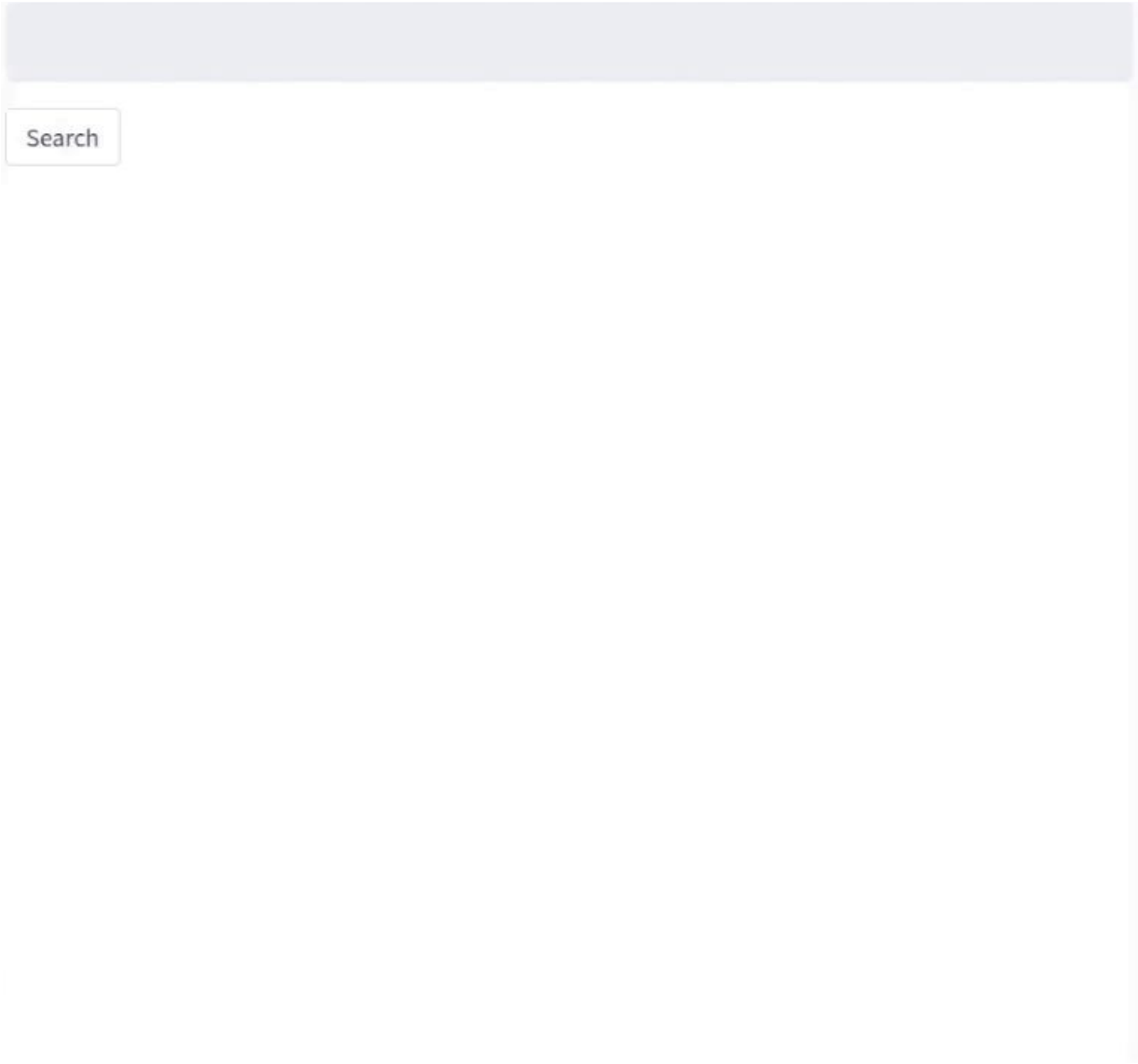
Valutazione sentence embedding con **classificatore KNN** su dataset ad hoc

Valutazione



	F1	Precision	Recall
Avg	0,83	0,84	0,82
IDF	0,85	0,86	0,84
SIF	**0,88**	**0,88**	**0,88**

Conclusioni



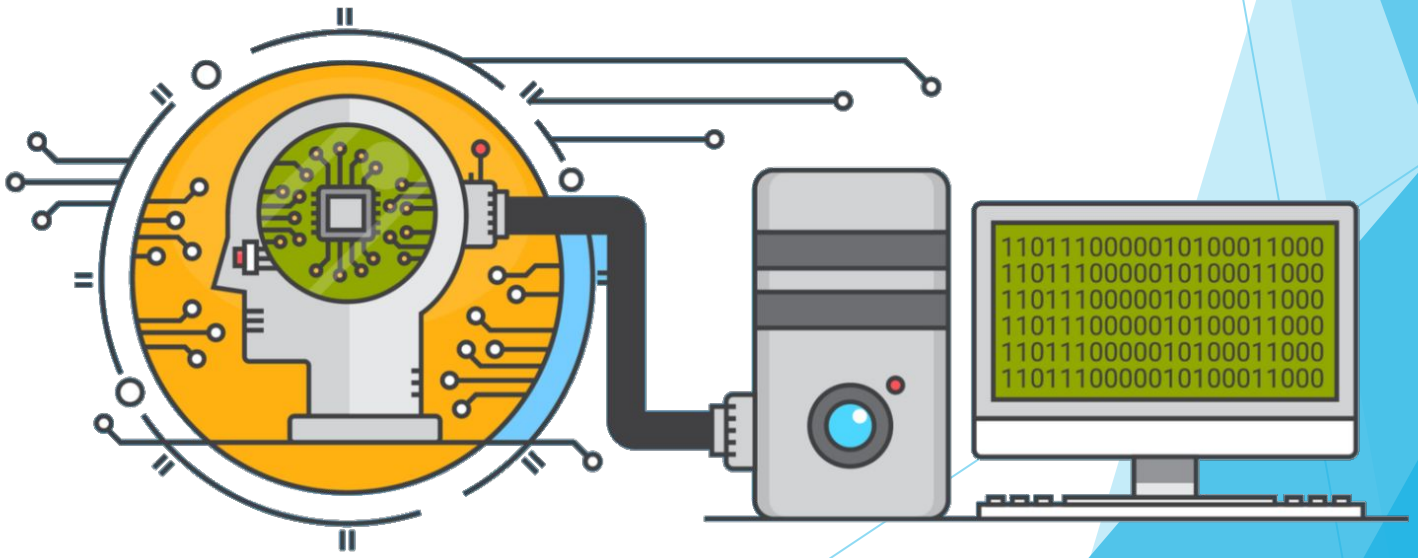
Risultati inerenti al
testo ricercato

Alta pertinenza dei risultati

Sviluppi futuri

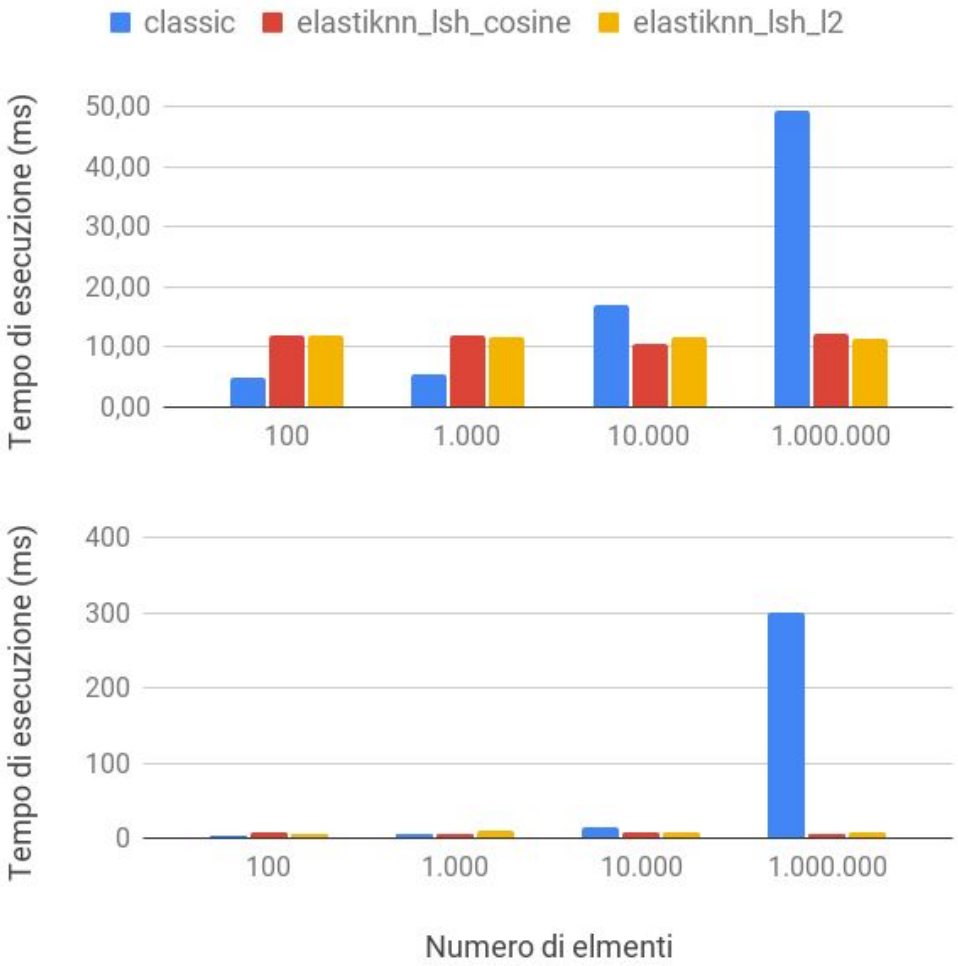
Affinamento dei risultati:

- Raccolta dati attraverso Feedback Implicito
- Re-ranking utilizzando reti neurali *semi-supervised*



Grazie per l’attenzione

Approximate Nearest Neighbors



46.851 elementi

300.000 elementi

Utilizza LSH per massimizzare collisioni tra hash e raggruppare embedding simili in bucket vicini

Ricerca non completa ma veloce